

Overview of Instruction tuning

Subject: Instruction tuning

1.

Measures In information theory, the concept of entropy is intricately linked to perplexity, a relationship notably established by Claude Shannon. Due to their ability to accurately predict the next token, LLMs are highly capable in lossless compression. A 2023 study by DeepMind showed that the model Chinchilla, despite being trained primarily on text, was able to compress ImageNet to 43% of its size, beating PNG with 58%.

2.

In 2020, OpenAI researchers demonstrated that their new model GPT-3 could understand what format to use given a few rounds of Q and A (or other type of task) in the input data as example, thanks in part due to the RLHF technique. This technique, called few-shot prompting, allows LLMs to be adapted to any task without requiring fine-tuning. Also in 2022, it was found that the base GPT-3 model can generate an instruction based on user input. The generated instruction along with user input is then used as input to another instance of the model under a "Instruction: [...], Input: [...], Output:" format. The other instance is able to complete the output and often produces the correct answer in doing so. The ability to "self-instruct" makes LLMs able to bootstrap themselves toward a correct answer.

3.

Stereotyping AI models can reinforce a wide range of stereotypes due to generalization, including those based on gender, ethnicity, age, nationality, religion, or occupation. When replacing human representatives, this can lead to outputs that homogenize or generalize groups of people. In 2023, LLMs assigned roles and characteristics based on traditional gender norms. For example, models might associate nurses or secretaries predominantly with women and engineers or CEOs with men due to the frequency of these associations in documented reality.

4.

Content filtering LLM applications accessible to the public, like ChatGPT or Claude, typically incorporate safety measures designed to filter out harmful content. However, implementing these controls effectively has proven challenging. For instance, a 2023 study proposed a method for circumventing LLM safety systems. In 2025, The American Sunlight Project, a non-profit, published a study showing evidence that the so-called Pravda network, a pro-Russia propaganda aggregator, was strategically placing web content through mass publication and duplication with the intention of biasing LLM outputs. The American Sunlight Project coined this technique "LLM grooming", and pointed to it as a new tool of weaponizing AI to spread disinformation and harmful content. Similarly, Yongge Wang illustrated in 2024 how a potential criminal could potentially bypass GPT-4o's safety controls to obtain information on establishing a drug trafficking operation. External filters, circuit breakers and overrides have been posed as solutions.

5.

Human provenance In 2023, Nature Biomedical Engineering wrote that "it is no longer possible to accurately distinguish" human-written text from text created by large language models, and that "It is all but certain that general-purpose large language models will rapidly proliferate... It is a rather safe bet that they will change many industries over time." Brinkmann et al. (2023) also argue that LLMs are transforming processes of cultural evolution by shaping processes of variation, transmission, and selection. As of October 2025, these early claims have yet to transpire and several HBR reports surface questions on the impact of AI on productivity.

6.

CBRN and content misuse AI labs treat CBRN defense (chemical, biological, radiological, and nuclear defense) and similar topics as high-consequence misuse attempt to apply various techniques to reduce potential harms. Some commenters expressed concern over accidental or deliberate creation of misinformation, or other forms of misuse. For example, the availability of large language models could reduce the skill level required to commit bioterrorism; biosecurity researcher Kevin Esvelt has suggested that LLM creators should exclude from their training data papers on creating or enhancing pathogens.

7.

reported arithmetics decoding the International Phonetic Alphabet unscrambling a word's letters disambiguating word-in-context datasets converting spatial words cardinal directions (for example, replying "northeast" in response to a 3x3 grid of 8 zeros and a 1 in the top-right), color terms represented in text. chain-of-thought prompting: In a 2022 research paper, chain-of-thought prompting only improved the performance for models that had at least 62B parameters. Smaller models perform better when prompted to answer immediately, without chain of thought. identifying offensive content in paragraphs of Hinglish (a combination of Hindi and English), and generating a similar English equivalent of Kiswahili proverbs. Schaeffer et al. argue that the emergent abilities are not unpredictably acquired, but predictably acquired according to a smooth scaling law. The authors considered a toy statistical model of an LLM solving multiple-choice questions, and showed that this statistical model, modified to account for other types of tasks, applies to these tasks as well. Let x be the number of parameter count, and y be the performance of the model.

8.

Tokenization also compresses the datasets. Because LLMs generally require input to be an array that is not jagged, the shorter texts must be "padded" until they match the length of the longest one.

9.

Sentience Contemporary AI practitioners generally agree that present-day large language models do not exhibit sentience. A minority view argues that even if there is a small chance that a given software system can have subjective experience, which some philosophers suggest is possible, then ethical considerations around potential large-scale suffering in AI systems may need to be taken seriously—similar to considerations given to animal welfare. Proponents of this view have proposed various precautionary measures like moratoriums on AI development and induced amnesia to address these ethical concerns. Leonard Dung argues that the evidential frameworks used to assess consciousness in animals apply equally to AI systems and that there is a significant probability near-future AI will be capable of suffering, making AI suffering risk a serious near-term ethical

concern that requires systematic mitigation. On the other hand, some existential philosophers argue there is no generally accepted way to determine if an LLM is conscious, given the inherent difficulty of measuring subjective experience. The 2022 Google LaMDA incident, where engineer Blake Lemoine claimed that the model was conscious, highlighted how LLMs can convince users that they are sentient through responses that do not prove sentience. Google described the engineer's claims as unfounded, and he was dismissed. Murray Shanahan argues that anthropomorphic framing of LLM capabilities encourages unwarranted attribution of cognitive properties to systems that operate through statistical pattern completion. Kristina Šekrst develops this further, arguing that LLMs function as "illusion engines" capable of producing outputs that coherently simulate properties such as consciousness without possessing them, but highlighting that, due to sophisticated creativity-temperature tradeoff, we may never be certain whether we are dealing with the emergence of consciousness or just a hallucination. David Chalmers similarly argues that while current LLMs likely lack features considered necessary for consciousness, extended successors incorporating these elements could plausibly meet the criteria within a decade.